

Readme

How to run the code

```
usage: main.py [-h] [--n N] [--net NET] [--buffer BUFFER] [--batch BATCH] [--gamma GAMMA] [--lr LR]
               [--eps_start EPS_START] [--eps_decay EPS_DECAY] [--eps_final EPS_FINAL] [--episode EPISODE]
               [--update_episode UPDATE_EPISODE] [--HER HER] [--HER_param HER_PARAM] [--HER_my_version HER_MY_VERSION]
               [--group GROUP] [--reward_version REWARD_VERSION] [--Curriculum CURRICULUM] [--interval INTERVAL]
               [--unfreeze_interval UNFREEZE_INTERVAL] [--freeze FREEZE] [--drop DROP] [--attention ATTENTION]
               [--emb_dims EMB_DIMS] [--head_num HEAD_NUM] [--cpu CPU] [--cuda_device CUDA_DEVICE]
               [--model_path MODEL_PATH] [--save_path SAVE_PATH] [--log_path LOG_PATH] [--train TRAIN]
               [--auto_test AUTO_TEST] [--test_nums TEST_NUMS]

DQN for Bits Sequence Generation

optional arguments:
  -h, --help            show this help message and exit
  --n N                Sequence Length
  --net NET             Network Hidden Layers' Dimension
  --buffer BUFFER       Buffer Size
  --batch BATCH         Batch Size
  --gamma GAMMA         Discount Factor
  --lr LR              Learning Rate
  --eps_start EPS_START
  --eps_decay EPS_DECAY
  --eps_final EPS_FINAL
  --episode EPISODE
  --update_episode UPDATE_EPISODE
                        Amount of Episodes to Update Network
  --HER HER            Whether to Use HER
  --HER_param HER_PARAM
  --HER_my_version HER_MY_VERSION
                        Whether to Use HER (my version)
  --group GROUP        Amount of Tokens to Generate at Once
  --reward_version REWARD_VERSION
                        Type of Reward Function (Refer to README for details.)
  --Curriculum CURRICULUM
                        Whether to Use Curriculum Learning
  --interval INTERVAL  Curriculum Interval
  --unfreeze_interval UNFREEZE_INTERVAL
                        Number of Curriculums to Unfreeze Parameters
  --freeze FREEZE      Initial Freezing Proportion of Curriculum Learning
  --drop DROP          Dropout Rate
  --attention ATTENTION
                        Whether to Add an Attention Layer to Network
  --emb_dims EMB_DIMS  Embedding Dimensions (before the Attention Layer)
  --head_num HEAD_NUM  Attention Head Number
  --cpu CPU            Use CPU
  --cuda_device CUDA_DEVICE
                        CUDA Device ID
  --model_path MODEL_PATH
                        Path to Load Model
  --save_path SAVE_PATH
                        Path to Save Model
  --log_path LOG_PATH  Path to Save Log
  --train TRAIN
  --auto_test AUTO_TEST
                        You can choose to test by hand or automatically.
  --test_nums TEST_NUMS
                        How many times you want to test the model.
```

You can utilize the aforementioned parameters in the main.py file for various models or tasks. Additionally, you can refer to the default values of these parameters by accessing the get_args function in main.py.

The five types of Reward Function are:

Reward Function0: $R_0(i) := (o_t == o_s)$

Reward Function1: $R_1(i) := (i == n) * similarity(o_t, o_s)$

Reward Function2: $R_2(i) := similarity(o_t, o_s)$

Reward Function3: $R_3(i) := \frac{o_t[i] == o_s[i]}{n}$

Reward Function4: $R_4(i) := \begin{cases} -1 & o_t[1:i-1] == o_s[1:i-1] \& o_t[i] \neq o_s[i] \\ (o_t == o_s) & otherwise \end{cases}$

Some Examples

Run Original DQN

```
1 | python main.py --n 10 --reward_version 1 --attention True --group 1
```

In this example:

- `n` represents the length of the target.
- `reward_version` represents the type of reward function.
- `attention` indicates whether to use the attention layer.
- `group` specifies how many tokens to generate in one step.

Run HER

```
1 | python main.py --n 10 --HER True
```

In this example, the flag `HER` determines whether to use HER.

Run Curriculum Learning

```
1 | python main.py --n 10 --Curriculum True
```

In this example, the flag `Curriculum` determines whether to use Curriculum Learning.

Run Test

To facilitate testing the model, a parameter file called `./n_35.npy` is provided in the current path. If you wish to use this parameter file, please refrain from changing any parameters in the code.

Automatic Test

```
1 | python main.py --n 35 --Curriculum True --train False --model_path ./n_35.npy
```

By setting the `train` parameter to False, the model will automatically run 100 times.

You can specify the path of the parameter file by setting the `model_path` parameter.

